

# Machine Learning for Natural Language Understanding

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The background image shows a workspace with a dark surface. A white notebook is open, showing a pencil sketch of a building. A yellow ruler is placed over the notebook. A black pen and a yellow pencil are visible. A small white tag with the word 'LEARN' is attached to a string. A hand is holding a black pen, ready to write.

# Learning

## Chapter 02.1

# Sources Lecture and Exercise

- Optimization (Derivatives)
  - <https://youtu.be/E5hU0AesA-I>
- Gradient Descent
  - <https://youtu.be/B3w5Zzuqi-E>

# Flipped Classroom Questions 1/2

- In which direction does the derivative point?



# Evaluation

## Chapter 02.2

# Sources Lecture and Exercise

- Hold-out data
  - <https://youtu.be/kFaoSM-wn4k>
- Overfitting / Underfitting
  - <https://youtu.be/x-OxfOHPLsc>
- Workflow (Train vs Validation vs Test)
  - [https://youtu.be/xAo77BqF\\_nI](https://youtu.be/xAo77BqF_nI)
- (Deep Dive: Model evaluation
  - <https://arxiv.org/pdf/1811.12808.pdf>
- Evaluation (simple metrics for binary classification)
  - <https://youtu.be/CTQm8T6PZKg>
- BLEU
  - <https://youtu.be/M05L1DhFqcw?si=ZOjNn62HYADD6Jh2>
- Perplexity
  - [https://youtu.be/NURcDHhYe98?si=T4jzZ9\\_Sk2Z4TC54](https://youtu.be/NURcDHhYe98?si=T4jzZ9_Sk2Z4TC54)

## Flipped Classroom Questions 2/2

- Why should I split my labeled data?
- What is overfitting? Name one potential reason for overfitting.
- How is the Tuning Set also called?
- What are elements of a confusion matrix?
- What does a probabilistic classifier produce?
- What are limits of binary classification metrics when applied in NLP?

“The future of language is about the development of language models...”<sup>1</sup>

1

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